

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0510

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 10%

Code of Conduct:

I Premarubha N (192525020) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Premarubha N

Rubrics for Evaluation

<i>Criteria</i>	<i>Weightage</i>	<i>Level 4 – Exemplary</i>	<i>Level 3 – Proficient</i>	<i>Level 2 – Developing</i>	<i>Level 1 – Beginning</i>	<i>Marks(100)</i>
<i>Understanding of Concepts</i>	<i>25%</i>	<i>Demonstrates thorough, accurate understanding of concepts</i>	<i>Good understanding with minor gaps</i>	<i>Basic understanding with some misconceptions</i>	<i>Poor understanding; major misconceptions</i>	
<i>Explanation</i>	<i>25%</i>	<i>Detailed, well-explained answers with relevant examples</i>	<i>Clear explanation with adequate details</i>	<i>Limited explanation; lacks depth</i>	<i>Poor or unclear explanation</i>	
<i>Application</i>	<i>20%</i>	<i>Effectively applies concepts with strong analytical ability</i>	<i>Applies concepts with minor errors</i>	<i>Limited application with weak analysis</i>	<i>No application or incorrect analysis</i>	
<i>Organization</i>	<i>15%</i>	<i>Well-structured answers with logical flow and coherence</i>	<i>Organized with minor issues in flow</i>	<i>Basic structure, lacks clarity</i>	<i>Poor organization, incoherent answers</i>	
<i>Accuracy</i>	<i>10%</i>	<i>Completely accurate and covers all required points</i>	<i>Mostly accurate with minor omissions</i>	<i>Partially correct with missing points</i>	<i>Incorrect answers with major gaps</i>	

<i>Clarity</i>	<i>5%</i>	<i>Clear, neat, professional presentation</i>	<i>Understandable with minor issues</i>	<i>Limited clarity, readability issues</i>	<i>Poorly written, difficult to understand</i>	
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1. An image appears too bright.

Explain and justify the most suitable grey level transformation to improve visibility, clearly describing the concept, application process, and expected outcome.

2. A dark image lacks detail.

Describe and justify an appropriate grey level transformation to enhance visibility, explaining how it improves low-intensity regions with proper reasoning.

3. An image has poor contrast.

Explain a suitable histogram-based technique for contrast enhancement and illustrate how it improves image quality with logical reasoning.

4. A low-contrast image requires uniform intensity distribution.

Describe and justify the method you would apply, explaining the concept, working process, and its effect on intensity levels.

5. An image shows uneven brightness across regions.

Explain and apply an appropriate enhancement technique to correct non-uniform illumination, providing a structured and accurate explanation.

6. A noisy image contains random intensity variations.

Explain and justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction.

7. An image requires noise reduction without significant loss of details.

Describe and apply an appropriate filtering approach, explaining how it balances noise removal and detail preservation.

8. An image appears blurred after smoothing.

Explain and justify a suitable sharpening method to restore details, including its working principle and expected results.

9. Fine details in an image are not clearly visible.

Describe and justify the filtering technique used to enhance details, explaining its effect on high-frequency components.

10. A noisy image must be preprocessed before further analysis.

Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification.

ANSWERS:

1. An image appears too bright.

Grey Level Transformation: Image Negative (or Intensity Reduction using Logically Inverted Gray Levels)

Concept

A bright image contains high intensity values. Applying the **Image Negative transformation** converts bright pixels into dark pixels, making overexposed regions easier to observe.

Transformation

$$s = L - 1 - r$$

Where:

- r = Input pixel intensity
- s = Output pixel intensity
- L = Total gray levels (256 for an 8-bit image)

Application Process

1. Read the image.
2. Apply the negative transformation to every pixel.
3. Display the transformed image.

Expected Outcome

- Bright regions become darker.
- Hidden details in bright areas become visible.
- Overall visibility improves.

2. A dark image lacks detail.

Grey Level Transformation: Power-Law (Gamma) Transformation

Concept

A dark image mainly contains low-intensity pixels. Gamma correction with $\gamma < 1$ increases these intensity values, making dark regions brighter.

Formula

$$s = cr^\gamma$$

Where:

- c = Constant
- $\gamma < 1$

Application Process

1. Normalize pixel values.
2. Apply gamma correction.
3. Convert back to the original intensity range.

Expected Outcome

- Dark areas become brighter.
- Hidden details become visible.
- Image visibility improves without excessive brightness.

3. An image has poor contrast.

Histogram-Based Technique: Histogram Equalization

Concept

Histogram Equalization redistributes pixel intensity values over the full gray-level range to improve contrast.

Working Process

1. Calculate the histogram.
2. Compute the cumulative distribution function (CDF).
3. Map old intensity values to new values.
4. Generate the enhanced image.

Expected Outcome

- Better contrast.
- More visible details.
- Wider intensity distribution.

4. A low-contrast image requires uniform intensity distribution.

Method: Histogram Equalization

Concept

Histogram Equalization spreads pixel intensities uniformly across the available gray levels.

Working Process

1. Compute image histogram.
2. Calculate cumulative histogram.
3. Transform intensity values using the cumulative function.
4. Produce an enhanced image.

Effect on Intensity Levels

- Pixel values become evenly distributed.
- Dynamic range increases.
- Contrast improves significantly.

5. An image shows uneven brightness across regions.

Enhancement Technique: Adaptive Histogram Equalization (CLAHE)

Concept

CLAHE enhances contrast locally rather than globally, correcting non-uniform illumination.

Working Process

1. Divide the image into small regions.
2. Apply histogram equalization to each region.
3. Limit contrast amplification.
4. Combine all regions smoothly.

Expected Outcome

- Uniform brightness.
- Better local contrast.
- Reduced over-enhancement.

6. A noisy image contains random intensity variations.

Smoothing Filter: Median Filter

Concept

A Median Filter removes impulse (salt-and-pepper) noise while preserving edges.

Working Mechanism

1. Select a neighbourhood window.
2. Arrange pixel values in ascending order.
3. Replace the centre pixel with the median value.

Impact

- Removes random noise.
- Preserves image edges.
- Produces smoother images.

7. An image requires noise reduction without significant loss of details.

Filtering Approach: Bilateral Filter

Concept

The Bilateral Filter smooths the image while preserving edges by considering both spatial distance and intensity difference.

Working Process

1. Compute spatial weights.
2. Compute intensity similarity weights.
3. Combine both weights.
4. Replace each pixel with the weighted average.

Expected Outcome

- Noise is reduced.
- Fine details are preserved.
- Edge sharpness remains intact.

8. An image appears blurred after smoothing.

Sharpening Method: Unsharp Masking

Concept

Unsharp Masking sharpens an image by adding high-frequency information back to the blurred image.

Working Principle

1. Blur the original image.
2. Subtract the blurred image from the original.
3. Add the difference back to the original image.

Expected Outcome

- Restores sharp edges.
- Improves clarity.
- Enhances fine details.

9. Fine details in an image are not clearly visible.

Filtering Technique: High-Pass Filter

Concept

A High-Pass Filter enhances edges and fine details by emphasizing high-frequency components.

Working Process

1. Apply a high-pass convolution kernel.
2. Suppress low-frequency components.
3. Enhance rapid intensity changes.

Effect on High-Frequency Components

- Edges become sharper.
- Fine textures become more visible.
- Overall image appears clearer.

10. A noisy image must be preprocessed before further analysis.

Spatial Filtering Method: Gaussian Filter

Concept

The Gaussian Filter smooths an image by reducing Gaussian noise before image processing.

Working Process

1. Create a Gaussian kernel.
2. Convolve the kernel with the image.
3. Replace each pixel with the weighted average of neighbouring pixels.

Justification

- Reduces random noise effectively.
- Produces smooth intensity transitions.
- Improves image quality for segmentation, feature extraction, and object detection.

Expected Outcome

- Reduced noise.
- Improved image quality.
- Better preprocessing for further analysis.